

Highly recoverable and robust rollable AMOLED display with smart elastomer materials

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Abstract

A new highly recoverable and robust AMOLED display with a novel double-stacked structure consisting of two smart elastomer layers with high modulus and low modulus was suggested and significantly improved pen drop height from 2 cm to 9.5 cm with additional special treatment for elastomer layer removing electro-static damage on panel image, coming from the increase of gate voltage of 1.2 V due to repeated friction between top layer and bottom layer during cyclic rolling. The top protective layer of elastomer enormously reduced the strain of the panel and PSA and therefore, lessened the deformation height of the rollable display at the edge up to 4 mm after a rolling radius of 15 mm and rolling cycles of 200,000, which looks almost flat. The top protective layer of elastomer with tensile hardening property was able to remarkably increase the energy damping effect for the pen drop test and led to the rise of pen drop height. The unique rolling strain distribution was systematically studied using a strain gauge sensor and simulation tool and validated by experimental results. This novel double-stacked structure also showed outstanding environmental reliability results as well.

KEYWORDS

flexible, island structure, organic ILD, TFT

1 | INTRODUCTION

Recently, active matrix organic light-emitting diodes (AMOLEDs) rollable display has received lots of attention as one of the next-generation future displays to embody ultimate expandability thanks to excellent portability. There are, however, some crucial features to be required for the development of rollable displays such as high flexibility, outstanding recoverability, and superb robustness in terms of panel and module structure. Besides these essential properties, rollable displays must show remarkable reliability under diverse environmental test conditions. But it has never been an easy task to meet this high robustness, brilliant flexibility, sufficient recoverability,

and superior reliability simultaneously with regards to the product of foldable, slidable, and rollable display.^{1–15}

We suggested several new impressive approaches to considerably enhance the recoverability, robustness, and anti-static reliability of the panel and module based on theoretical analyses of strain distribution, anti-shock phenomena, and anti-static tests.

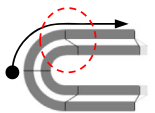
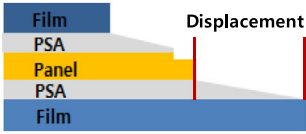
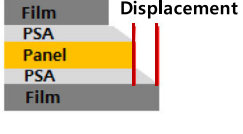
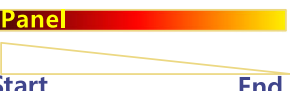
2 | EXPERIMENTS

In rollable display, pressure sensitive adhesive (PSA) located between each brittle layer with high modulus usually undergoes severe shear strain due to the very low

modulus of pressure sensitive adhesive (PSA) used in order to reduce the strain of hard layer with high modulus and therefore, minimize the total strain of module structure. This maximizes the shear strain of pressure sensitive adhesive (PSA) at the edge of the rollable display panel which is much higher than the shear strain at the edge of the foldable display panel since the rollable display experiences accumulative strain all over the panel and therefore, has a maximum shear strain at the end of panel during rolling process, unlike foldable display that

suffers localized strain in folding area. To compare shear strain distribution in rolling and folding states, we used a simple stacked structure where PSA was located between the dummy panel (PI) and protective film (PI), and each of two protective films was placed over and under panel respectively. With rolling radius of 10 mm and folding radius of 10 mm, PSA in rollable display had shear strain of 1,000% ~ 1,500% but PSA in foldable display had much lower shear strain of less than 400% in Figure 1.

FIGURE 1 Shear strain distribution in rolling and folding with a simple stacked structure of rolling radius and folding radius of 10 mm where PSA is located between dummy panel and protective film. Each of the two protective films is placed over and under the dummy panel, respectively.

R10	Rolling	Folding
Form		
Displacement	 Shear Strain: 1000~1500%	 Shear Strain: < 400%
Shear Strain Distribution Pattern	 Start End	 Center

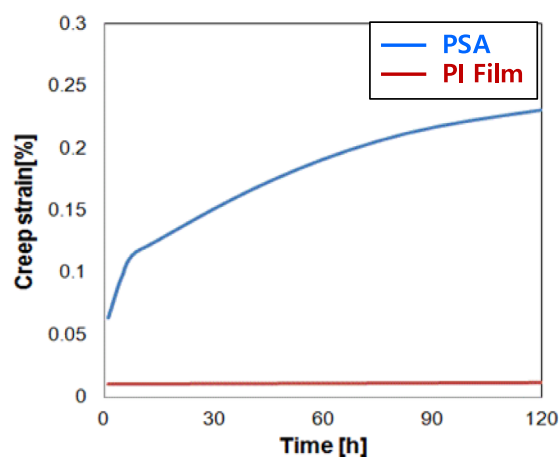
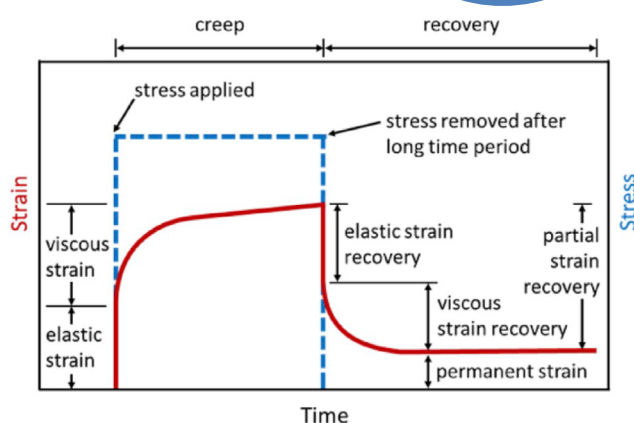
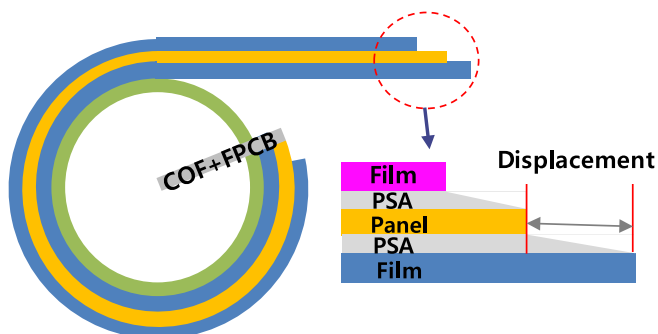
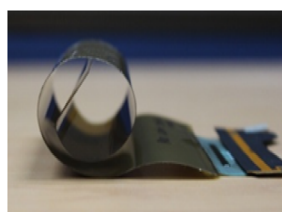


FIGURE 2 Permanent deformation at the edge of the panel after static rolling (rolled) condition due to creep property of PSA.

Pressure sensitive adhesive (PSA) has an innate property of creep under a constant stress condition which leads to permanent non-reversible deformation of panel and module after static and dynamic rolling in Figure 2.¹⁶ Therefore, it is very hard to fully restore panel and module to a initial flat state after static rolling and cyclic rolling stress. To improve the recovery performance of rollable display, highly restorable films with low modulus such as elastomer including thermoplastic polyurethane (TPU) with Young's modulus of 27 MPa, polydimethylsiloxane (PDMS) with Young's modulus of 3 MPa and so on, might be very desirable in Figure 3.

The simple module structure with polyimide (PI) as a dummy panel instead of real panel, employing several elastomers and polyimide as protective layer, was used at top side and bottom side of the module structure to compare recovery property of each film. Elastomer films demonstrated remarkable recoverability after rolled test of 10 hours at rolling radius of 5 mm and reduced deformation height from 51 mm (polyimide) to 19 mm (coating

PU), 9 mm (TPU), and 3 mm (PDMS) due to excellent resilient property of these elastomer materials in Figure 3. When comparing the recovery performance of polyimide (film widely used in flexible display module) and elastomer based on flexible panel, elastomer layer showed much better recovery performance than polyimide layer after rolling cycles of 5,000 for polyimide and rolling cycles of 200,000 for elastomer at rolling radius of 15 mm, respectively. Elastomer layer reduced deformation height at the edge of panel and module from 35 mm to 4 mm even if rolling cycles are increased from 5,000 to 200,000 in Figure 4.

A simple module stacked structure consisting of three polyimide layers and two PSA layers was used in order to analyze shear strain distribution pattern of PSA during the rolled state. To estimate the shift of PSA after rolling and rolled test, the initial positions of PSA and PI were marked using laser beam, and displacements of initially laser-marked positions of PSA were calculated using microscope after rolling or rolled test. Simulation results

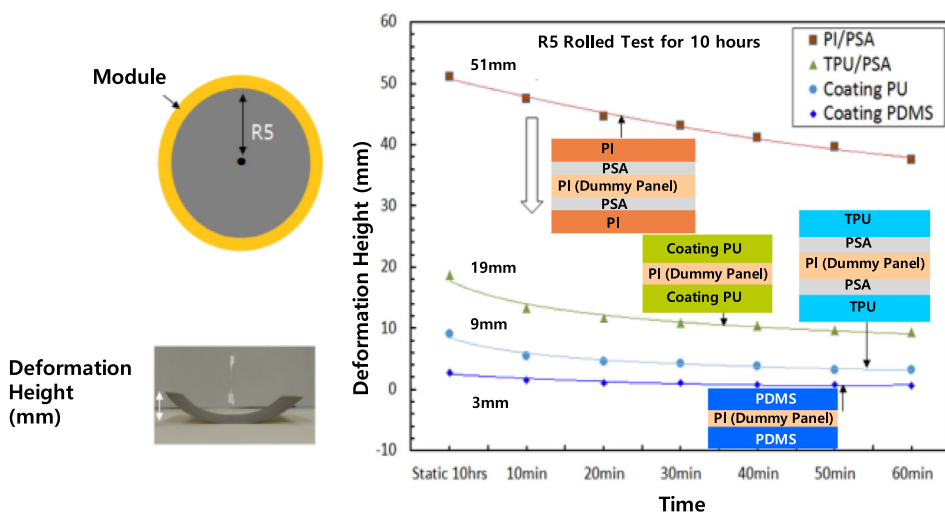


FIGURE 3 Deformation height at the edge of simple module structure after rolled test for 10 hours at the rolling radius of 10 mm for different films of PI, coating PU, TPU, and PDMS.

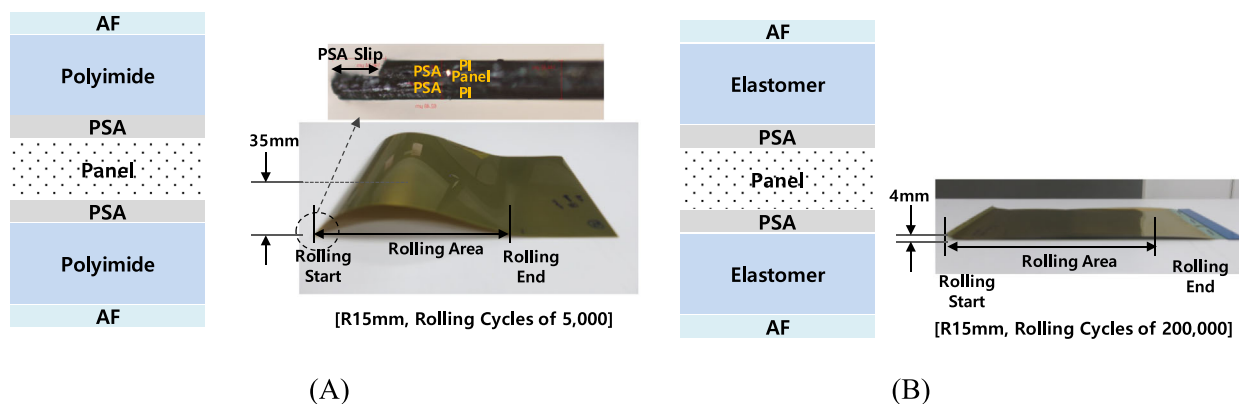


FIGURE 4 Deformation height result of simple module structure after rolling cycles of (A) 5,000 for polyimide film and (B) rolling cycles of 200,000 for elastomer film at the rolling radius of 15 mm.

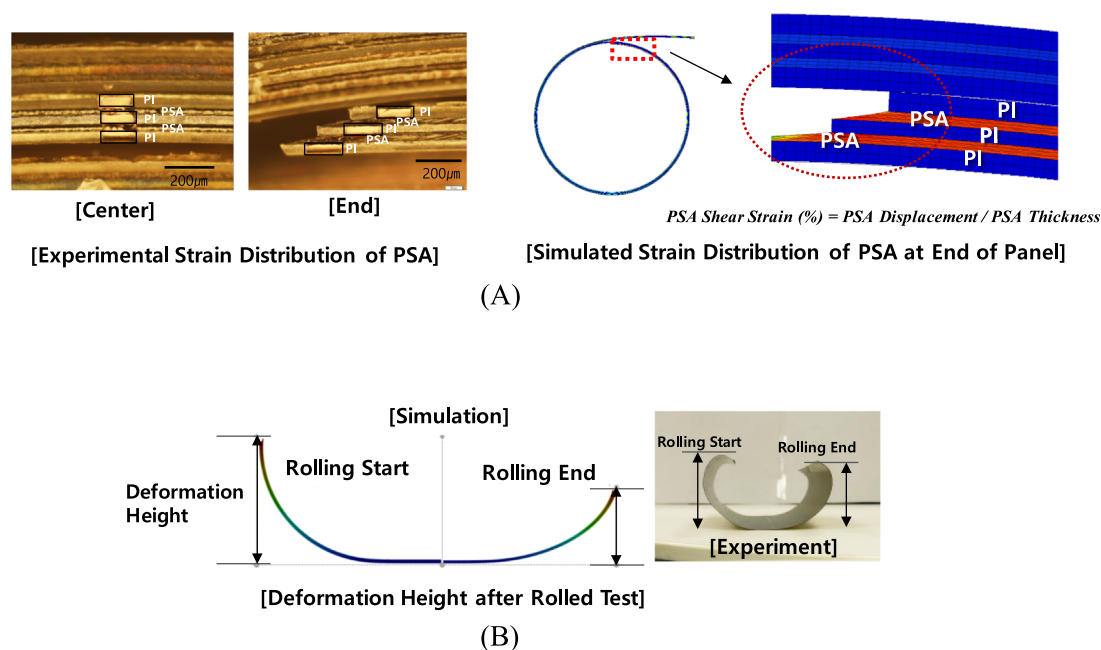


FIGURE 5 (A) Results of shear strain distribution of PSA after rolled state for 24 hours at rolling radius of 10 mm obtained by experiment and simulation (ABAQUS) and (B) results of deformation height of panel and module after rolled test for 1 hour at rolling radius of 5 mm obtained by experiment and simulation (ABAQUS).

obtained by ABAQUS and the experimental result indicated similar shear strain distribution of PSA after rolled state for 24 hours at rolling radius of 10 mm and shear strain of PSA continued to increase toward the end of panel and module in Figure 5A. Simulation results obtained from ABAQUS and experimental result presented that larger deformation height occurred at rolling start position than rolling end position after rolled state for 1 hour at rolling radius of 5 mm because higher shear strain happened at rolling start position coming from a smaller rolling radius at rolling start position in Figure 5B.

As rolling radius decreases, the position of maximum shear strain is shifted from the center of panel and module toward the position of rolling start due to the unique strain distribution of rolling in Figure 6A where the experimental test was carried out at the condition of rolled state for 24 hours at rolling radius of 5 mm, 10 mm, and 15 mm. During rolling process, the decoupling of strain in PSA layers to enable multiple neutral planes (NPs) to take place usually at both ends of panel and module in Figure 6B. The maximum shear strain of PSA exists near the center of panel and module, since displacement of PSA is almost zero at the center when rolling radius is large and displacement of PSA becomes dominant at the end of panel and module in Figure 6A and B. There is one neutral plane at the center of panel and module and the possibility of creating multiple neutral planes (NPs) increases toward the end of panel and

module, because shear strain of the PSA grows toward the end of panel and module in Figure 6B. Strains of PI layer at several positions of top and bottom layers of this module structure were measured using a strain sensor and these results of strains on PI layer is very similar to simulation results using simulation tool (ABAQUS) in Figure 6C. Simulation results showed that as the friction force increases between the top and bottom layer of PI during rolling process, shear zero point shifts toward the start position in Figure 6C. Simulation results of PI strain are in good accordance with experimental results of PI strain obtained by strain sensor.

Regarding rolling strain of polyimide (PI) film as a top protective layer, strain of PI film as a top protective layer has to be evaluated by considering two kinds of strain effect which include rolling strain of PI film when PI film is rolled and another additional strain reduction effect of PI film at the end of panel coming from occurrence of multiple NP due to PSA's decoupling (caused by large shear strain of PSA) at the end of panel in Figure 7A and Figure 6B. Therefore, we must take into account these two kinds of strain effects to influence PI film as a top protective layer during rolling process. When shear strain of PSA between PI film continues to increase toward the end of panel (module structure), it reduces strain of PI more and more on PSA. If multiple NP happens at the end of panel (module structure), strain of PI film as a top layer away from NP does not continue to increase during rolling process as moving up from the

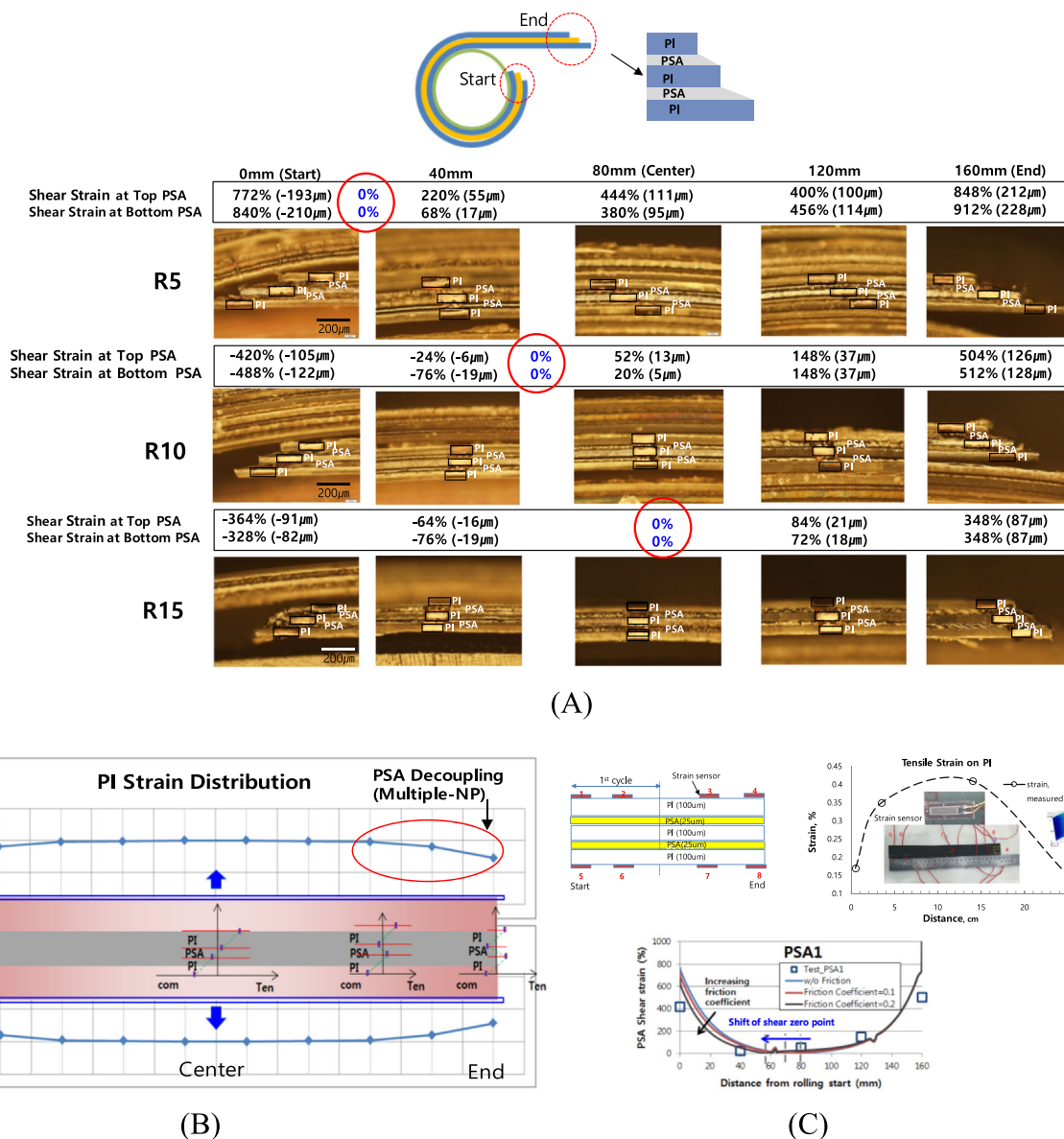


FIGURE 6 PSA strain at rolled state: (A) shear strain measured using microscope, (B) strain distribution of PI based on simulation analysis, and (C) strain of PI measured using strain sensor ($\square$) and shear strain of PSA evaluated by simulation tool (ABAQUS) after rolled state for 24 hours at rolling radius of 10 mm.

panel, but strain of PI film as a top layer is reset to the initial strain and strain of PI film is almost similar to strain of layer when rolling only PI layer. If we take a close look at strain distribution of module structure of rollable display at the center and end position of panel in Figure 6B, as moving from bottom PI layer to the top PI layer, strain continues to increase at the center of panel, but strain of bottom PI layer increases and strain of top PI layer rather decreases at the end of panel (this is strain reduction effect of PI layer) because multiple NP happens due to large shear strain of PSA. Figure 7A and B shows that the position of the neutral plane does not exist at the center but is displaced away from the center due to

the added rolling strain as the radius of rolling gets smaller and smaller.

As the rolling radius decreases from 15 mm to 5 mm, the position of maximum strain of PI on the top of module structure is shifted from the center of panel and module toward the position of rolling start as shown from the experimental results measured by strain sensor in Figure 7B, since rolling strain is inversely proportional to rolling radius.

This explanation about a unique rolling strain distribution was also confirmed by experimental result in Figure 8, which showed that the position where display damages like crack or thin-film encapsulation (TFE)

FIGURE 7 Rolling strain of PI located on the top of simple module structure during rolling process: (A) concept for rolling strain distribution of PI and (B) rolling strain of PI measured by strain sensor.

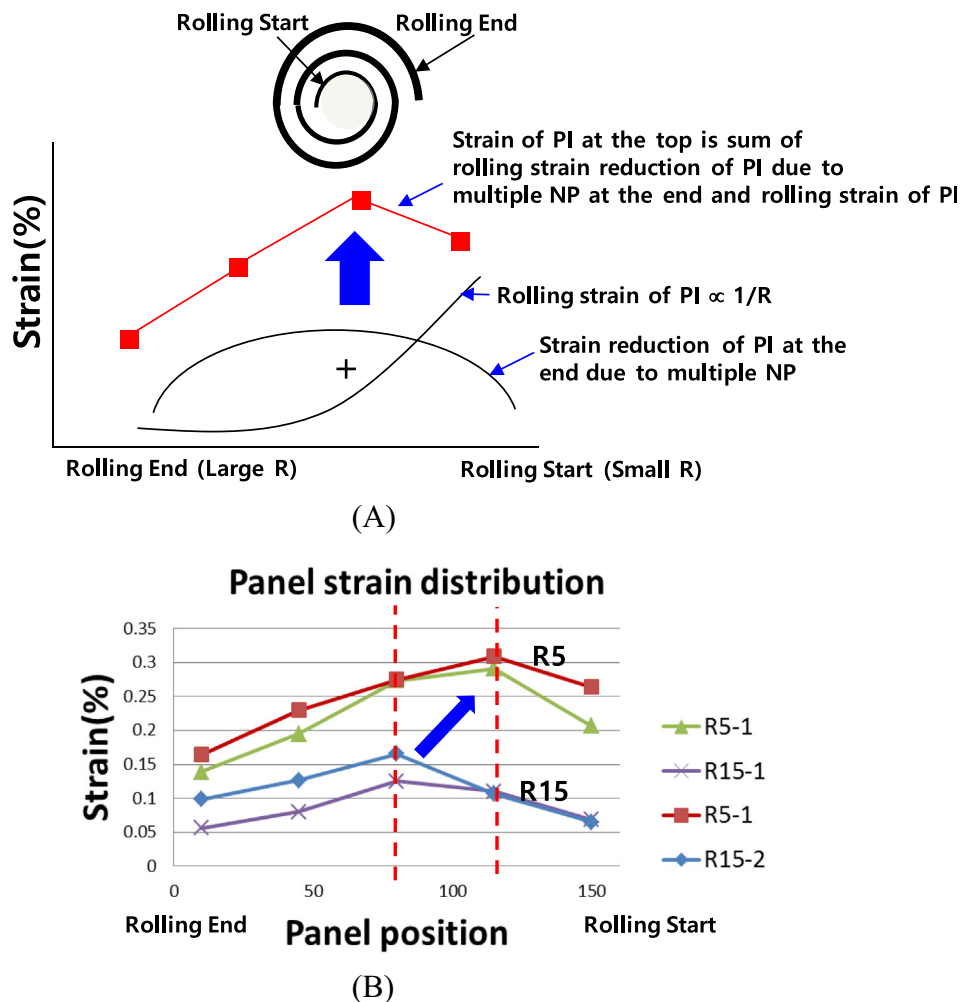
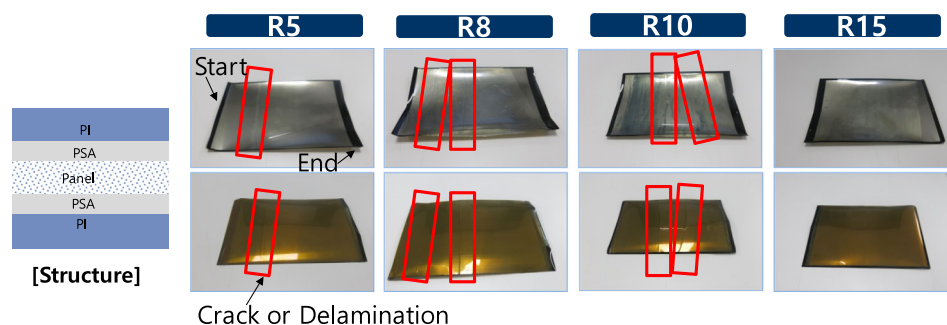


FIGURE 8 Position of crack of backplane or delamination of thin-film encapsulation (TFE) in panel during rolling process with different rolling radius.



delamination happened due to the growth of accumulated strain, was shifted from the center of panel and module to the position rolling start as rolling radius becomes smaller. These damages such as cracks, buckling, and delamination took place first at the position of the maximum strain inside display.

In order to lessen rolling radius, the elastomer protective layer can be applied to rollable module structure thanks to its low modulus and high recoverability. When the polyimide (PI) layer on the top of simple module structure was replaced with elastomer layer, shear strain

of PSA at the end significantly declined from 420–512% to 36–104% due to low modulus and high flexibility of elastomer in Figure 9. Therefore, elastomer material can be strongly considered when rollable display with a small rolling radius is necessary.

Cyclic rolling usually causes panel damage due to electro-static discharge accumulated by repetitive contact between the top and bottom layer of the module structure as shown in Figure 10A. The electric field triggered by this electro-static discharge is at least higher than 10^7 V/m, which amounts to rise of gate voltage of 1.2 V

in Figure 10A. Anti-static treatment with sheet resistance (R_s) less than $10^8 \Omega/\square$ on the top and bottom layer film is able to get rid of the electro-static damage of panel in Figure 10B.

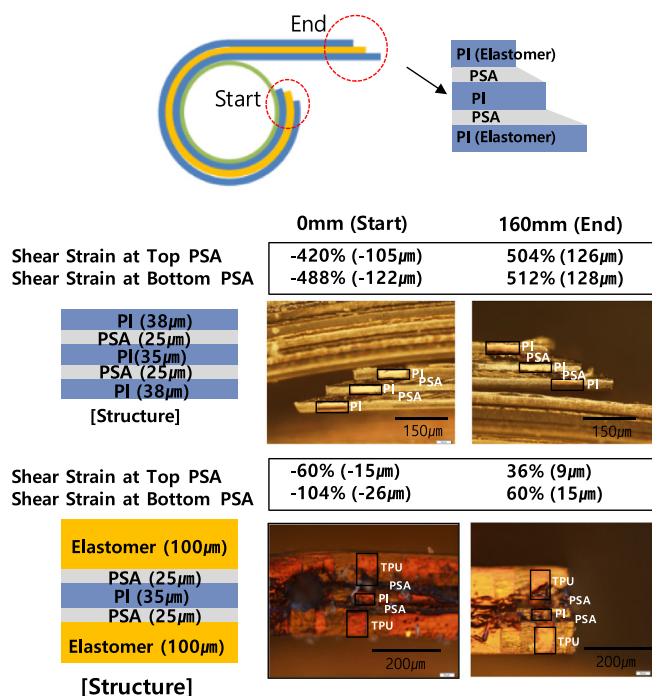


FIGURE 9 Total strain of PI and TPU on the top of module structure after rolled state for 24 hours at rolling radius of 10 mm.

It is indispensable that rollable panel must be robust to external impact such as pen drop and ball drop. Pen drop height is the maximum height where panel does not show any panel damage such as dark pixels and bright pixel. In the pen drop test, diameter of tip of pen ranges from 0.3 mm to 0.7 mm, and panel (or module) is placed on the plate. In order to enhance pen drop height, a new robust stacked structure including two layers with different moduli, was suggested in Figure 11, where a layer with a high modulus had to be placed as close to panel as possible to prevent delamination or buckling during cyclic rolling, since positioning layer with high modulus as close to panel as possible (neutral plane [NP] is normally located inside panel) is able to lessen large stress (σ) of hard layer with high modulus by diminishing distance (L) between layer with high modulus and neutral plane (NP) during rolling process. L is proportional to strain (ϵ) of layer with high modulus as shown in equation (1) and equation (2), where r is rolling radius. This new structure consisted of two separate layer (two separate functions), which located elastomer layer with low modulus and high flexibility at the top layer of module structure to effectively absorb external impact energy and placed hard layer with high modulus just above and below panel to reduce the strain of panel as much as possible in a collision with a falling ball. This novel structure showed an outstanding increase in pen drop height from 2 cm to 9.5 cm in Figure 11A. As the modulus of

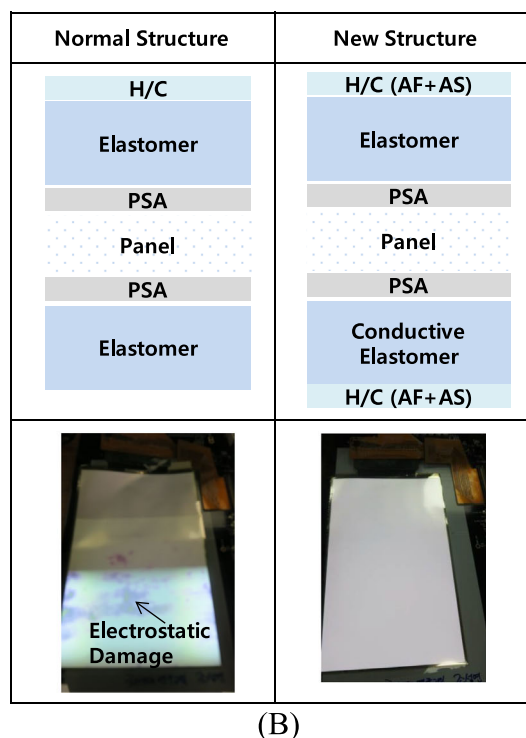
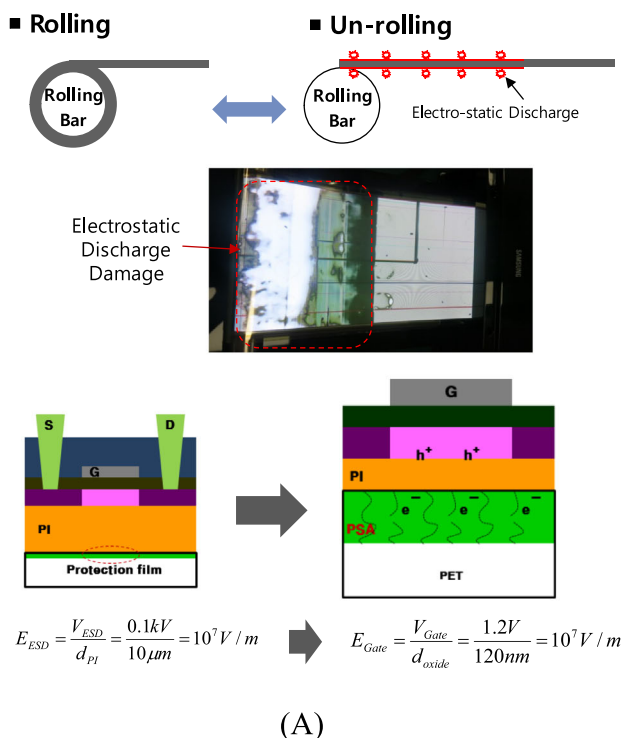
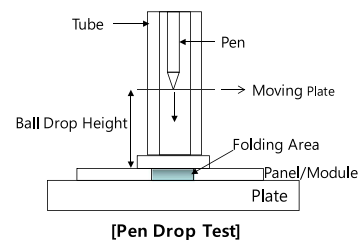
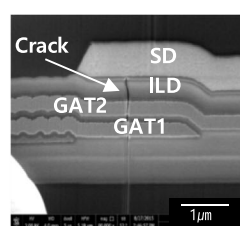
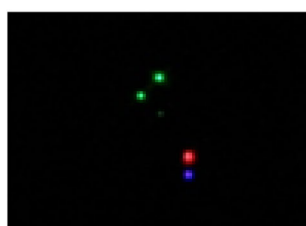
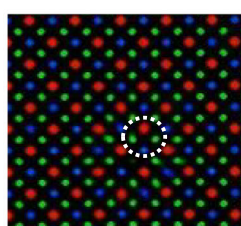


FIGURE 10 (A) Panel damage caused by electrostatic discharge and electric field induced by electrostatic discharge, and (B) anti-static treatment effect on panel of rollable display.

	Normal	High Robust Structure-I	High Robust Structure-II
Structure			
Pen Drop Height (cm)	2cm	6.5cm	9.5cm

(A)



(B)

FIGURE 11 (A) Robust stacked structure and (B) image of bright pixel, dark pixel, and crack in pen drop impact.

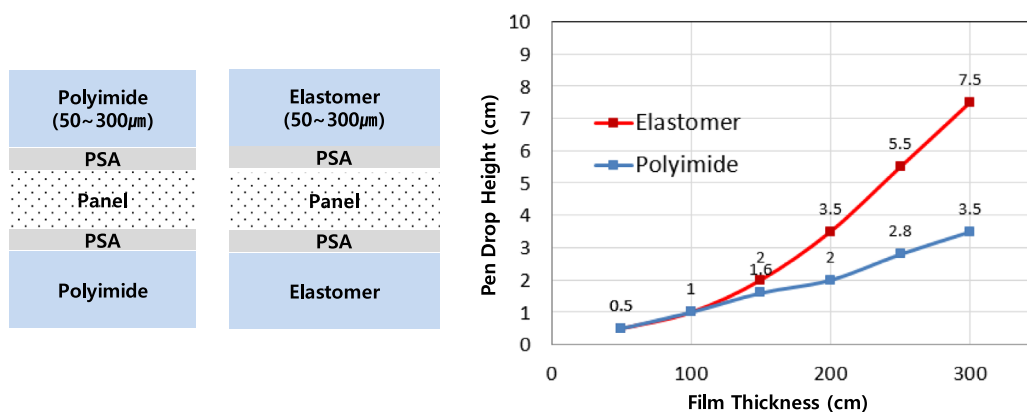


FIGURE 12 Pen drop test results for different thicknesses of polyimide and elastomer.

hard layer increases, the pen drop height rises due to less strain on panel.¹⁷ However, growth of modulus of hard layer also deteriorates flexibility of module structure. There is a trade-off between robustness and rollability. In this unique module structure, elastomer layer plays a role in improving external energy damping and flexibility and the hard layer with high modulus plays a role in lessening strain of backplane and panel. When the strain of panel exceeds a critical strain in a collision with falling ball, dark pixel or bright pixel occurs in Figure 11B.

As thickness of elastomer protective film and polyimide protective film increased from 50 μ m to 300 μ m, pen drop height for elastomer protective film increased exponentially but pen drop height for polyimide mounted linearly in Figure 12. Elastomer is formed as a unit structure of hard segment and soft segment in Figure 13B. For elastomer protective film, Young's modulus abruptly rose at some tensile rate as the tensile rate continued to increase in Figure 13A. The reason for sharp increase of Young's modulus is supposed to come from tensile thickening effect of elastomer (similar to the shear thickening effect in viscoelastic material), which might be caused by enhancing the alignment of hard segment parts of elastomer as tensile rate increases.^{18,19} As thickness of elastomer increases, tensile rate where Young's modulus of elastomer sharply rises, decreases. We supposed that this early occurrence of tensile thickening could improve the effectiveness of external impact energy

damping when pen drop test was performed, since in ball drop test, the tensile rate of elastomer is the highest when ball first collides with elastomer and the tensile rate of elastomer gradually decreases as falling ball presses

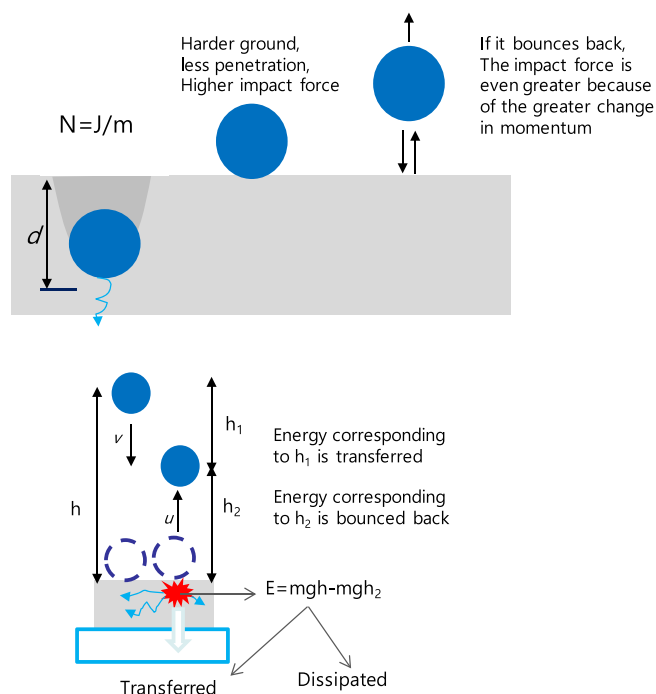
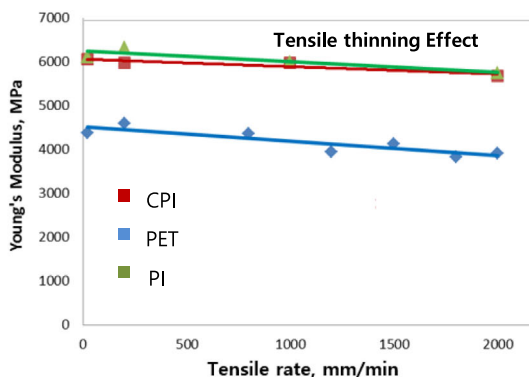
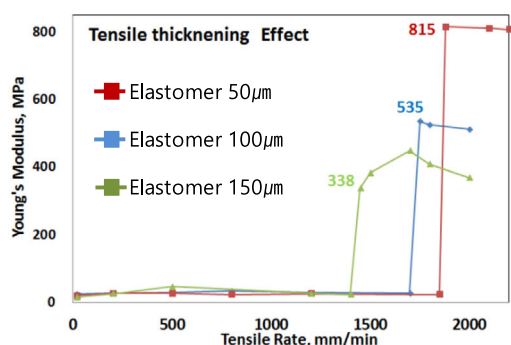
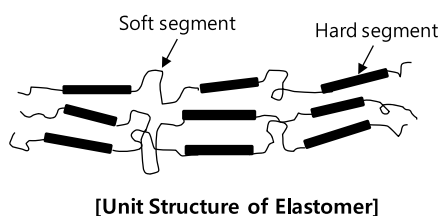


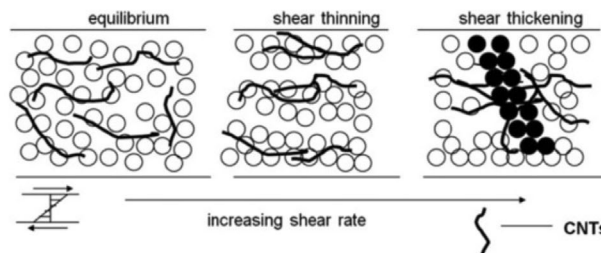
FIGURE 14 Energy damping process for ball drop test.



(A)



(B)



(C)

FIGURE 13 (A) Young's modulus vs. tensile rate for different thicknesses of elastomer and CPI (colorless polyimide), PET, PI, (B) typical unit structure of elastomer, and (C) shear thickening phenomenon for viscoelastic materials.

deeper into elastomer. As the thickness of elastomer increases, the tensile rate of elastomer at which the modulus of elastomer sharply rises, declines (tensile thickening happens early). Eventually, it is assumed that as the thickness of the elastomer grows, the time for tensile thickening to operate, becomes longer, and thus energy damping effect on the external impact of falling ball upsizes.

We also suppose that at the time of the occurrence of shear thickening of elastomer, too high an increase

of Young's modulus of elastomer could reduce the effectiveness of external impact energy damping in a collision of ball because elastomer becomes too hard. For PI and pol poly-ethylene terephthalate (PET) materials, however, Young's modulus gradually declined as tensile rate increases, which is a typical phenomenon for plastic materials, called as tensile thinning (similar to shear thinning in viscoelastic materials) in Figure 13C. Shear thickening effect indicates that viscosity increases due to increased alignment inside viscoelastic material as shear rate goes up and shear thinning means that viscosity decreases as shear rate rises in viscoelastic material.

When ball (or pen) falls on panel and module, energy damping layer of elastomer can first reduce the external impact energy by absorbing external impact energy and the ball presses energy damping layer into a certain depth (d), called a penetrating depth in equation (3) and (4), and transfers remaining impact energy due to penetration depth (d) and contact time (dt) toward high modulus layer in Figure 14. The hard layer with high modulus plays a role of ramping down level of panel bending caused by transferred impact energy and therefore, avoiding crack of panel by maintaining strain of the panel below critical strain where the crack of panel occurs.

This extremely robust and highly recoverable rollable AMOLED display was also very stable after outward rolling cycles of 20,000 at rolling radius of 5 mm and after 240 hours at high temperature and high humidity condition of 85°C/85% RH, and low-temperature condition of -40°C, respectively without any noticeable deterioration of image quality in Figure 15.

We also demonstrated a proto-type 12.4-in. automotive rollable display for CID (Center

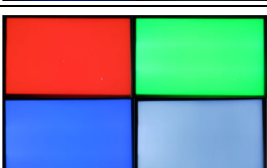
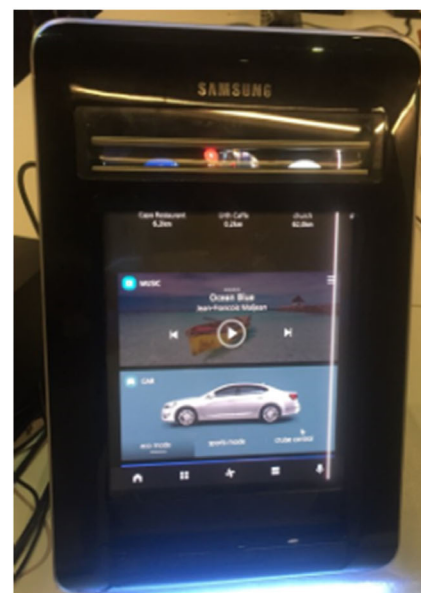
Environmental Reliability Test Condition	Panel Image After Test
200,000 Rolling at 25°C	
85°C/85% for 240 hours	
-40°C for 240 hours	

FIGURE 15 Panel image after environmental test results such as rolling cycles of 200,000 at 25°C, 85°C/85% for 240 hours, and -40°C for 240 hours, respectively.



FIGURE 16 Proto-type 12.4-in. Automotive rollable display for CID (center information display) with rolling radius of 5 mm.



Information Display) with rolling radius of 5 mm. The unique advantage of rollable display can extremely expand the display screen by just rolling out from a small space in Figure 16.

3 | CONCLUSION

A remarkable robust stacked structure for rollable display using smart elastomer material was proposed based on systematic analysis of simulation and experiment. This new structure significantly enhanced the robustness and recoverability of the rollable display with two separate layer consisting of layers with high modulus and low modulus. Layer with low modulus played a role in improving external impact energy damping and the layer with high modulus played a role in lessening strain of panel. This novel stacked structure considerably elevated pen drop height from 2 cm to 9.5 cm and eliminated electro-static damage on panel during cyclic rolling by introducing anti-static treatment to the elastomer. This 12.4-in. the smart stacked structure of rollable AMOLED display is also very robust after outward rolling cycles of 20,000 at rolling radius of 5 mm and after 240 hours at high temperature and high humidity condition of 85°C/85% RH, and low-temperature condition of −40°C, respectively without any noticeable deterioration of image quality. We also demonstrated a proto-type 12.4-in. automotive rollable display for CID (Center Information Display) with the rolling radius of 5 mm.

([Equation])

$$\sigma = E\varepsilon \quad (1)$$

$$\varepsilon \sim \frac{L}{r} \quad (2)$$

$$F = \frac{mgh}{d} \quad (3)$$

$$F = m \frac{dv}{dt} = m \frac{[v - (-u)]}{dt} \quad (4)$$

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How to cite this article: Kim T, Choi J, Kim M, Kim H, Kim J, Kim S, et al. Highly recoverable and robust rollable AMOLED display with smart elastomer materials. *J Soc Inf Display*. 2025;33(5): 594–605. <https://doi.org/10.1002/jsid.2087>